

- 8** In an Argand diagram, the point P representing the complex number w lies on the locus defined by $\{z: \arg(z-7) = \frac{3}{4}\pi\}$. You are given that $\operatorname{Re}(w) = 1$.

(a) Find w . **[2]**

The point P also lies on the locus defined by $\{z: |z+3-9i| = k\}$, where k is a constant.

(b) Find the complex number represented by the other point of intersection of the loci defined by $\{z: |z+3-9i| = k\}$ and $\{z: \arg(z-7) = \frac{3}{4}\pi\}$. **[7]**